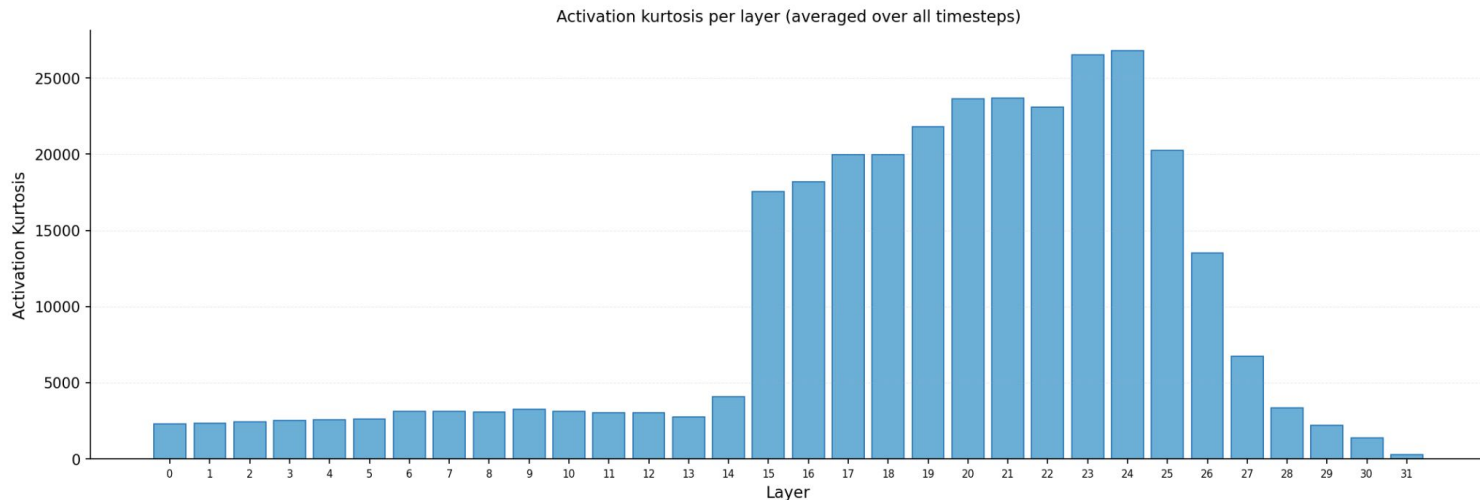


Kurtosis Analysis & QuaRot/DuQuant DLM Quantization

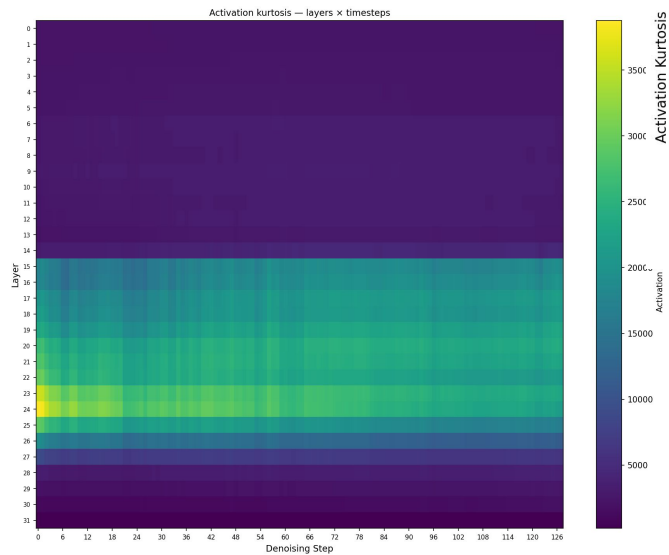
Ahmad Fraij

Kurtosis Analysis of LLaDa Model

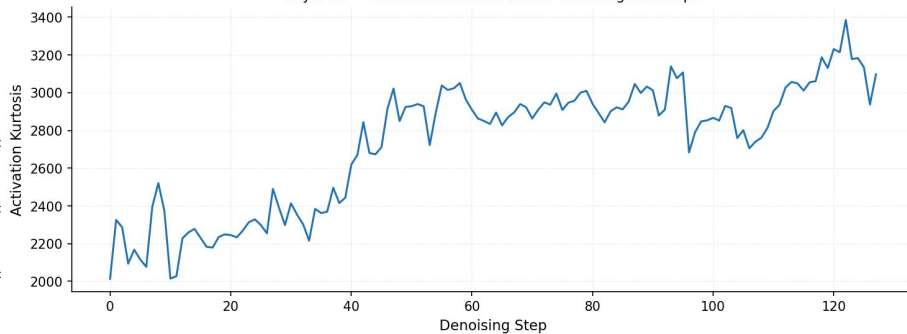
- High kurtosis and extreme outliers appear in the middle layers (15-25)
- These values are calculated by flattening the activation matrix for each layer and calculated mean across prompts and across all timesteps



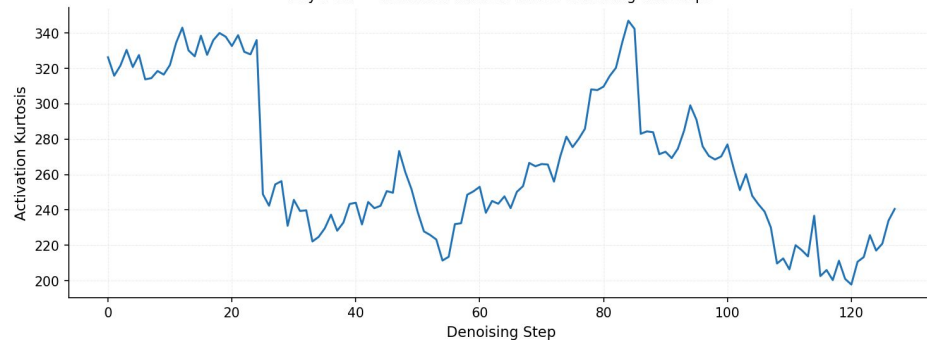
Kurtosis Analysis of LLaDa Model



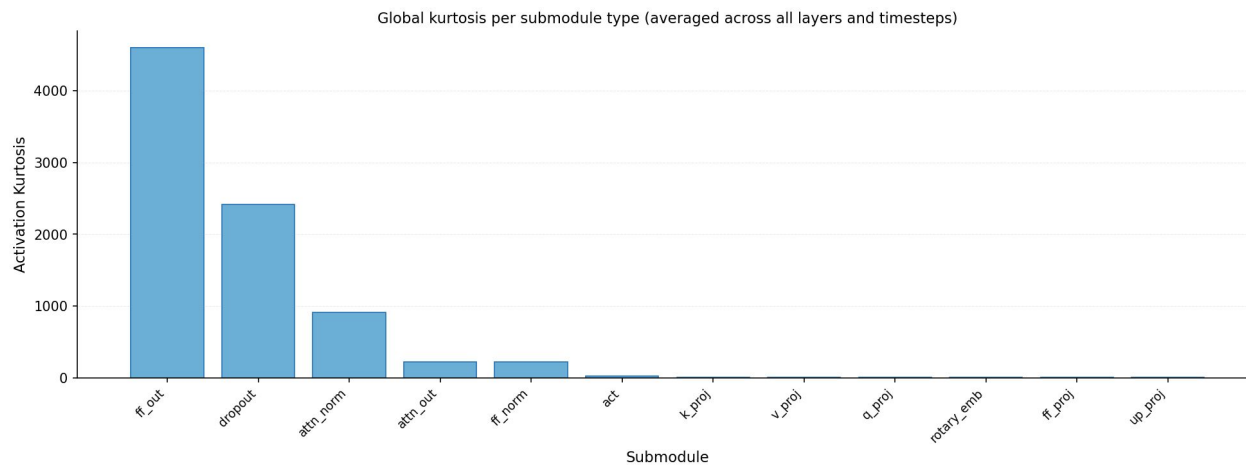
Layer 29 — activation kurtosis across denoising timesteps



Layer 31 — activation kurtosis across denoising timesteps



Kurtosis Analysis of LLaDa Model



SmoothQuant

- Uses a smoothing technique to shift quantization challenge from activations to weights
- Migrating errors to the weight matrix will increase weight quantization error, so a balance between both weight and activation using a hyperparameter alpha

$$\mathbf{s}_j = \max(|\mathbf{X}_j|)^\alpha / \max(|\mathbf{W}_j|)^{1-\alpha}$$

QuaRot

- Dealing with outliers in activations by rotating the activations using randomized Hadamard transformations
- QuaRot has two main stages:
 1. Manipulate model's weights in full precision and add two Hadamard transformations in the model's forward pass
 2. Quantize weights and add do online quantization of activations.
- GPTQ is used by default to quantize weights
- RTN is used for activation quantization

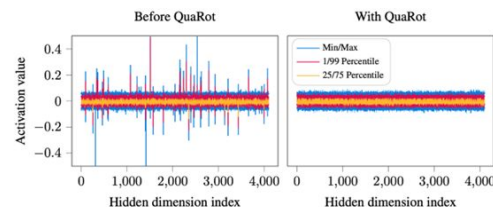


Figure 1: The distributions of activations at the input to the FFN block in LLAMA2-7B model, in the tenth layer. Left: using the default configuration as downloaded from Hugging Face. Right: after processing using QuaRot. The processed distribution has no outliers, leading to superior quantization.

DuQuant

- Redistributes massive and normal outliers across different channels using transformations and rotations
- Normal Outliers: persistent in several channels across all tokens
- Massive outliers: appear on smaller subset of tokens with exceedingly high values
- The two main parts are:
 - Local redistribution of feature outliers using block diagonal rotation matrices
 - Global redistribution of outliers across different blocks using zigzag permutation
- RTN is used for both weight and activation quantization

DuQuant vs QuaRot

- QuaRot uses fixed, data-independent Hadamard rotations while DuQuant uses data-aware (usually wikitext data), block-wise rotations plus permutation to also handle the massive outliers.

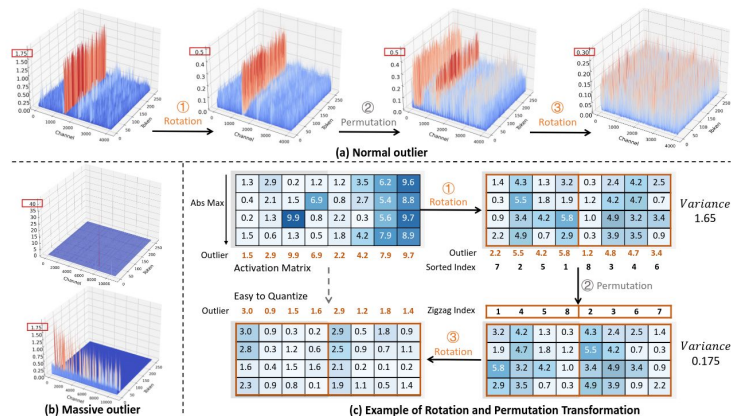


Figure 2: Transformation Steps for Activation Matrices after smooth technique. (a) Sequential transformations on Normal Outliers: ① initial rotation to reduce outliers within blocks, ② permutation to evenly distribute outliers across blocks, and ③ a second rotation for further smoothing. (b) Activation changes for Massive Outliers before and after DuQuant. (c) A sample matrix for highlighting the continual reduction of outliers through rotation and permutation, with outliers marked in dark blue.

QuaRot and DuQuant Evaluation

Weight-activation quantization of Dream-7B-Base and LLaDA-8B-Base. Results show flip rates between the base and quantized models. RTN is used as weight quantizer for quarot and duquant method

Model	Setting	Method	ARC-C	ARC-E	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base	W4A16	SINQ	3.58	3.49	1.69	6.47	8.60	1.65
	W4A8	Absmax (SINQ)	6.06	4.33	2.72	8.92	11.8	1.57
		QuaRot (RTN)	13.39	10.36	6.26	18.16	22.23	6.04
		DuQuant (RTN)	6.49	5.47	3.38	9.39	13.49	—
	W4A4	Absmax (SINQ)	34.72	44.36	29.87	47.04	51.69	28.67
		QuaRot (RTN)	16.21	14.82	8.97	25.89	28.81	8.67
		DuQuant (RTN)	10.24	8.84	4.90	16.42	22.29	—
Dream-7B-Base	W4A16	SINQ	6.23	3.41	3.37	5.68	—	—
	W4A8	Absmax (SINQ)	6.31	3.96	3.54	7.97	—	—
		QuaRot (RTN)	46.25	59.72	34.99	38.44	—	—
		DuQuant (RTN)	12.63	8.67	7.56	12.79	—	—
	W4A4	Absmax (SINQ)	46.33	60.86	33.08	37.97	—	—
		QuaRot (RTN)	47.19	61.49	34.82	36.78	—	—
		DuQuant (RTN)	45.65	58.72	32.27	34.41	—	—

Plan for Next Week

- Use SING/GPTQ/AWQ as weight quantizer for duquant and quarot and compare the results
- Analyze issues with these techniques